END-TO-END MLOps

Heart Disease Classification

Machine Learning Operations (MLOps) — AIMLCZG523

Assignment 01 | Total marks: 50

This report is generated from local artifacts available at build time. Any absent metric or screenshot is marked as unavailable rather than fabricated.

Health disclaimer: educational project only; not for diagnosis, treatment, triage, or clinical decisions.

1. Executive Summary

This project implements a reproducible binary classification workflow for the UCI Heart Disease processed Cleveland subset. It covers data acquisition, EDA, preprocessing, Logistic Regression and Random Forest model comparison, cross-validation and tuning, MLflow experiment tracking, model serialization, automated tests, CI configuration, FastAPI serving, Docker packaging, Kubernetes manifests, and Prometheus metrics.

The original UCI target `num` is converted to a binary target: 0 remains 0 and values above 0 become 1. The data is historical and small, so results do not establish clinical validity or generalization to other populations.

2. Runtime Status and Actual Results

Dataset rows: 303; predictor features: 13; class 0: 164; class 1: 139.

Missing values by column: {"age": 0, "ca": 4, "chol": 0, "cp": 0, "exang": 0, "fbs": 0, "oldpeak": 0, "restecg": 0, "sex": 0, "slope": 0, "target": 0, "thal": 2, "thalach": 0, "trestbps": 0}

Selected model: logistic_regression. Test-set size: 61.

Model	Accuracy	Precision	Recall	F1	ROC-AUC	CV ROC-AUC ± SD
logistic_regression	0.885	0.839	0.929	0.881	0.965	0.902 ± 0.015
random_forest	0.869	0.812	0.929	0.867	0.953	0.888 ± 0.032

3. Data Acquisition and EDA

The acquisition script downloads the processed Cleveland dataset from the UCI Machine Learning Repository when no local copy is present. The loader handles the original question-mark missing-value convention, normalizes numeric fields, removes duplicate rows, and derives the binary target. Missing feature values are imputed inside the model pipeline rather than before the split.

EDA checks include feature histograms, class balance, a correlation heatmap, and a JSON summary. Correlations describe association and do not imply causation.

Figure 1. Feature distributions

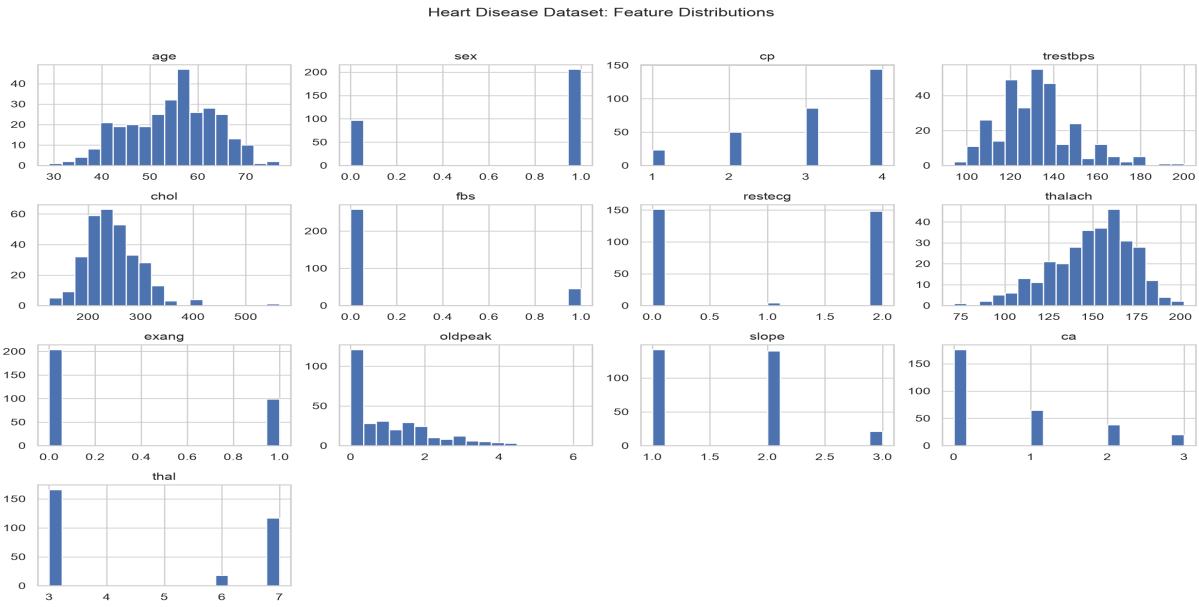


Figure 2. Binary class balance

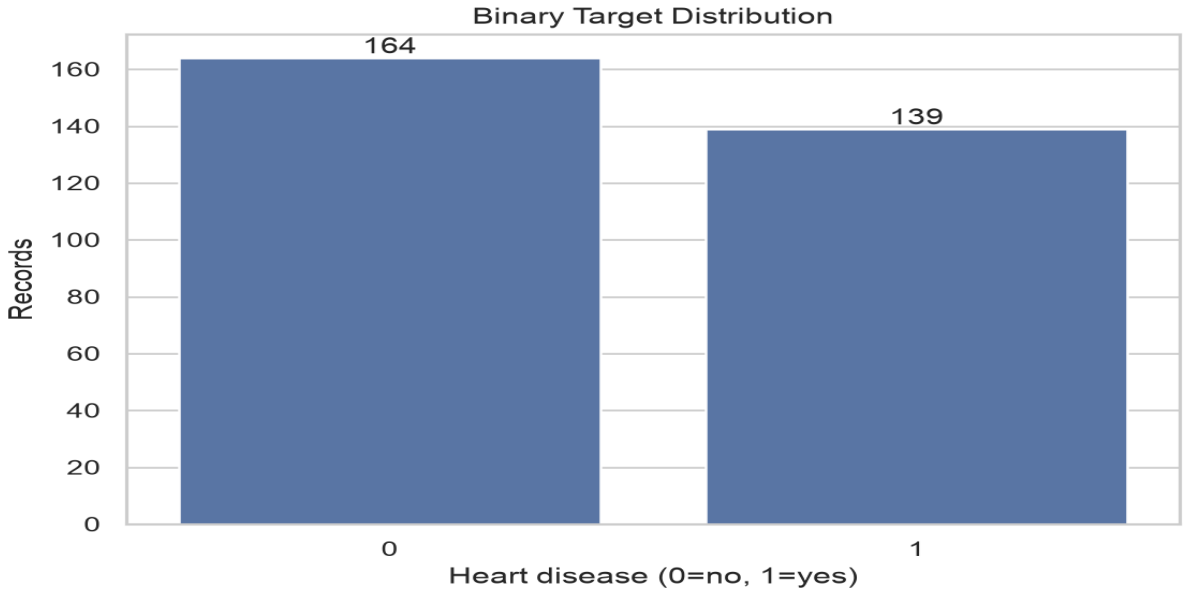
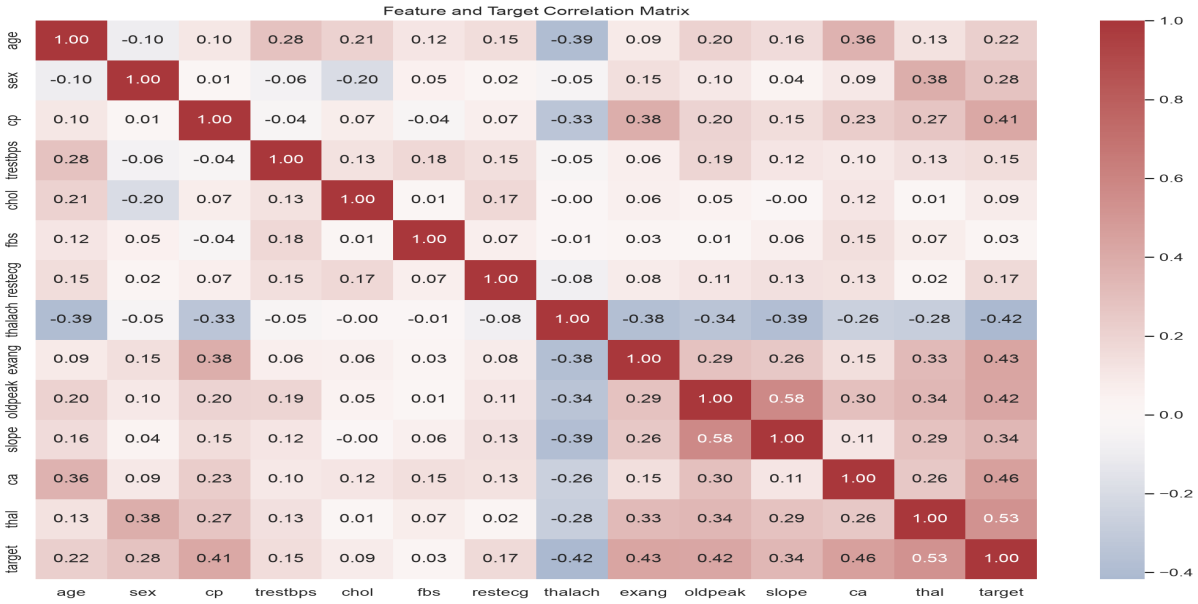


Figure 3. Feature/target correlation heatmap


Feature and Target Correlation Matrix
age 1.00 -0.10 0.10 0.28 0.21 0.12 0.15 -0.39 0.09 0.20 0.16 0.36 0.13 0.22
sex -0.10 1.00 0.01 -0.06 -0.20 0.05 0.02 -0.05 0.15 0.10 0.04 0.09 0.38 0.28
cp 0.10 0.01 1.00 -0.04 0.07 -0.04 0.07 -0.33 0.38 0.20 0.15 0.23 0.27 0.41
trestbps 0.28 -0.06 -0.04 1.00 0.13 0.18 0.15 -0.05 0.06 0.19 0.12 0.10 0.13 0.15
chol 0.21 -0.20 0.07 0.13 1.00 0.01 0.17 -0.00 0.06 0.05 -0.00 0.12 0.01 0.09
fbs 0.12 0.05 -0.04 0.18 0.01 1.00 0.07 -0.01 0.03 0.01 0.06 0.15 0.07 0.03
restecg 0.15 0.02 0.07 0.15 0.17 0.07 1.00 -0.08 0.08 0.11 0.13 0.13 0.02 0.17
thalach -0.39 -0.05 -0.33 -0.05 -0.00 -0.01 -0.08 1.00 -0.38 -0.34 -0.39 -0.26 -0.28 -0.42
exang 0.09 0.15 0.38 0.06 0.06 0.03 0.08 -0.38 1.00 0.29 0.26 0.15 0.33 0.43
oldpeak 0.20 0.10 0.20 0.19 0.05 0.01 0.11 -0.34 0.29 1.00 0.58 0.30 0.34 0.42
slope 0.16 0.04 0.15 0.12 -0.00 0.06 0.13 -0.39 0.26 0.58 1.00 0.11 0.29 0.34
ca 0.36 0.09 0.23 0.10 0.12 0.15 0.13 -0.26 0.15 0.30 0.11 1.00 0.26 0.46
thal 0.13 0.38 0.27 0.13 0.01 0.07 0.02 -0.28 0.33 0.34 0.29 0.26 1.00 0.53
target 0.22 0.28 0.41 0.15 0.09 0.03 0.17 -0.42 0.43 0.42 0.34 0.46 0.53 1.00
age sex cp trestbps chol fbs restecg thalach exang oldpeak slope ca thal target
1.0
0.8
0.6
0.4
0.2
0.0
−0.2
−0.4

4. Feature Engineering and Model Development

Numeric features are median-imputed and standardized. Categorical-coded attributes are imputed with the most frequent value and one-hot encoded with unknown values ignored. Both transformations are fitted as part of a scikit-learn pipeline, reducing data leakage during cross-validation.

Logistic Regression provides a linear baseline. Random Forest can model nonlinear patterns and feature interactions. Grid search uses five-fold StratifiedKFold and ROC-AUC scoring. Reported metrics include accuracy, precision, recall, F1-score and ROC-AUC, alongside cross-validation ROC-AUC mean and standard deviation.

Evaluation caveat: the compact implementation uses the held-out split to compare tuned candidates. A rigorous unbiased final estimate should reserve an untouched test set after model selection.

Figure 4. Selected model confusion matrix

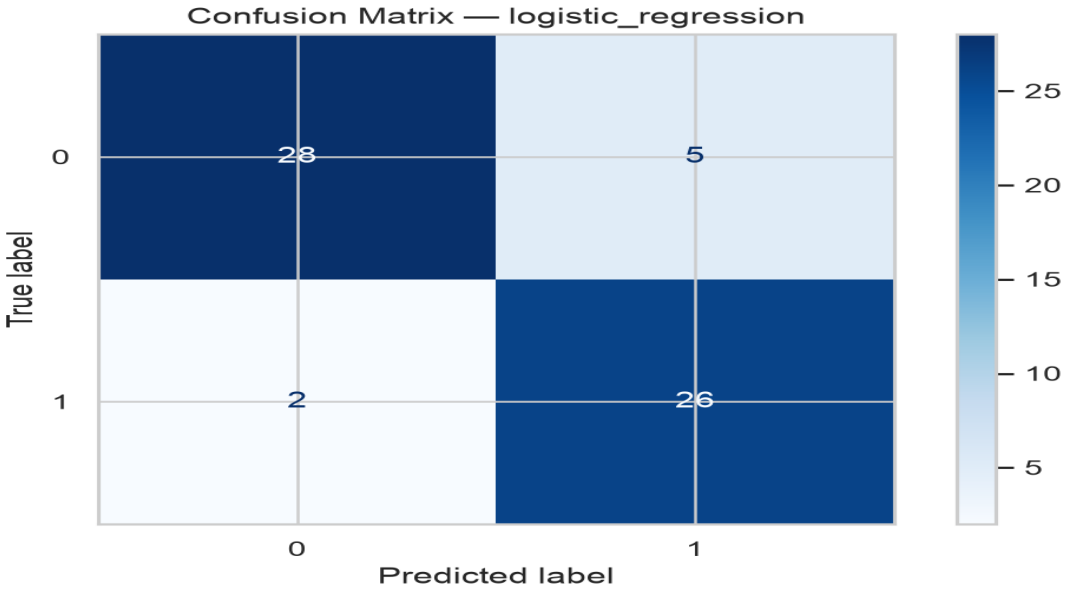
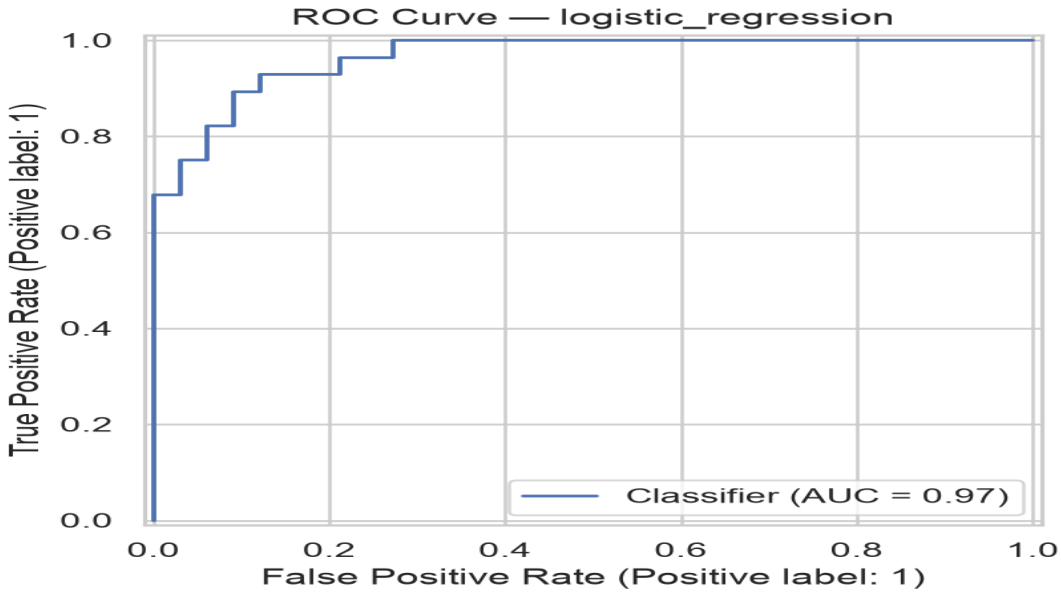


Figure 5. Selected model ROC curve



5. Experiment Tracking and Packaging

MLflow records model family, seed, split, fold count, cross-validation metric summaries, tuned parameters, held-out metrics, serialized sklearn model, and classification report. Start the local UI with `mlflow ui --backend-store-uri ./mlruns --host 127.0.0.1 --port 5000` and inspect both runs.

The selected preprocessing-plus-estimator pipeline is saved with joblib. A model card records the selected model, target definition, metrics, parameters and limitations. Keep dependency versions reproducible by freezing the successfully tested environment if the instructor requires exact package versions.

6. Automated Testing and CI/CD

Pytest verifies preprocessing behavior and basic API endpoints. Ruff checks code style. The GitHub Actions workflow installs dependencies, validates the dataset, runs tests and EDA, trains the models, builds a Docker image, and uploads available run artifacts. It runs on pushes, pull requests, and manual dispatch.

GitHub Actions evidence is unavailable unless the repository has been pushed to GitHub and the workflow has run. Do not claim a passing workflow without checking its actual status.

7. API and Containerization

FastAPI exposes `/`, `/health`, `/docs`, `/predict`, and `/metrics`. The prediction request schema validates the 13 required features and returns a binary prediction, label, predicted-class confidence and positive-class probability. These probabilities are model outputs, not calibrated clinical risk.

```
python -m src.heart_disease.train
uvicorn src.heart_disease.api:app --host 127.0.0.1 --port 8000
docker build -t heart-disease-api:latest .
docker run --rm -p 8000:8000 heart-disease-api:latest
```

The Docker image copies the source and serialized model into a Python 3.11 container. Build the image only after the model artifact exists. Test `/health`, `/docs`, `/predict` and `/metrics` on the running service.

8. Kubernetes Deployment

```
minikube start
docker build -t heart-disease-api:latest .
minikube image load heart-disease-api:latest
kubectl apply -f deployment/
kubectl rollout status deployment/heart-disease-api
kubectl get pods,svc
```

The manifest defines two replicas, resource requests/limits, liveness/readiness probes and a LoadBalancer Service. Local Minikube may require `minikube service heart-disease-api --url` or port forwarding. No public endpoint is claimed unless a remote cluster is actually configured.

9. Monitoring, Security, and Limitations

The API emits structured logs to stdout and exposes Prometheus counters for request outcome and a histogram for request duration. Prometheus can scrape `/metrics`; Grafana needs a configured Prometheus data source and dashboard. If those steps have not been run, the report records them as missing.

Before any real deployment, add authentication, TLS, rate limiting, network restrictions, secret management, log retention and vulnerability scanning. Do not log raw patient features or identifiable health information.

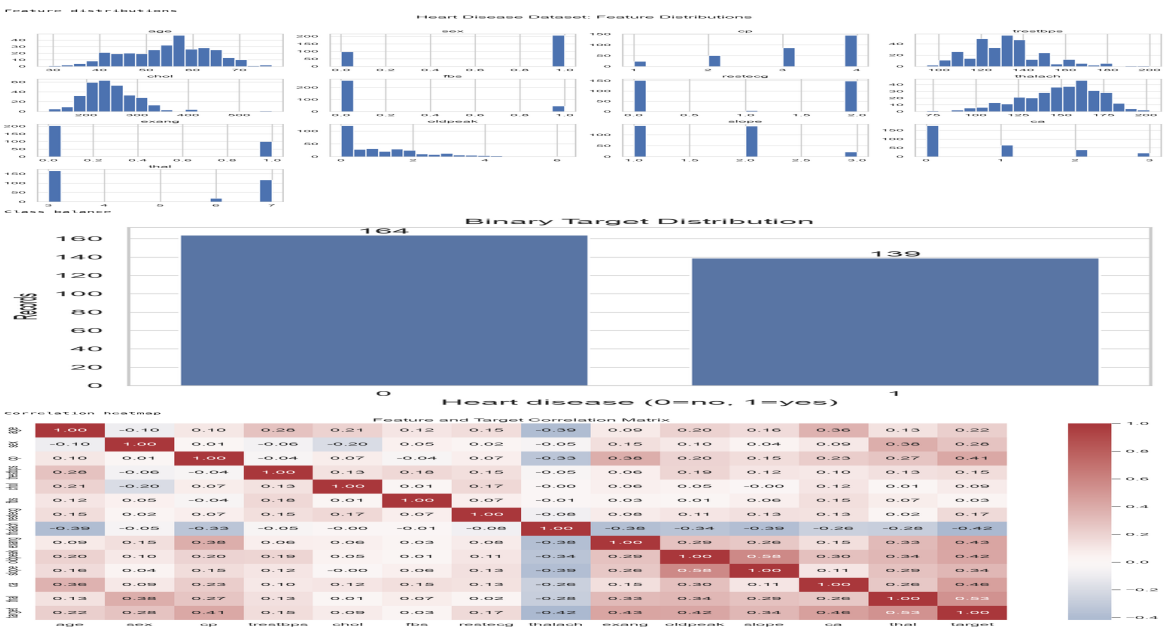
The dataset is small and historical. It cannot establish clinical validity, safety, fairness, or performance on current patient populations. The service is an educational demo only.

10. Evidence and Remaining Work

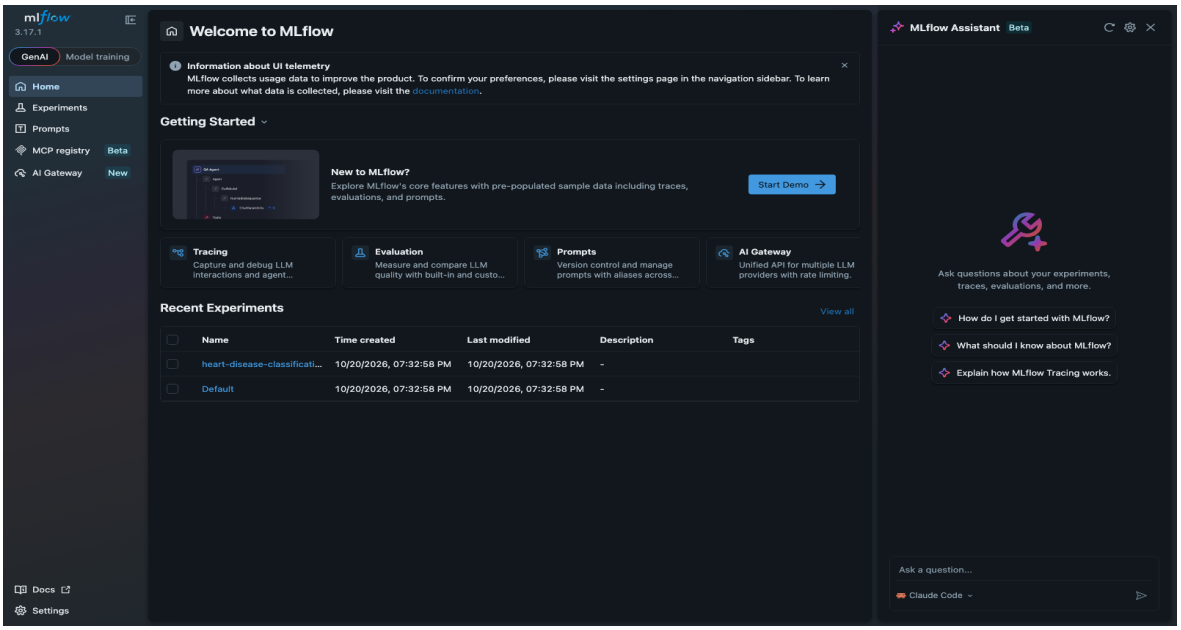
Evidence item	Status
dataset summary	Available
training summary	Available
comparison metrics	Available
serialized model	Available
test evidence screenshot	Available
docker evidence screenshot	MISSING
kubernetes evidence screenshot	MISSING
github actions evidence screenshot	MISSING

Missing screenshots: 05_docker_build.png, 08_kubernetes.png, 10_grafana.png, 11_github_actions.png

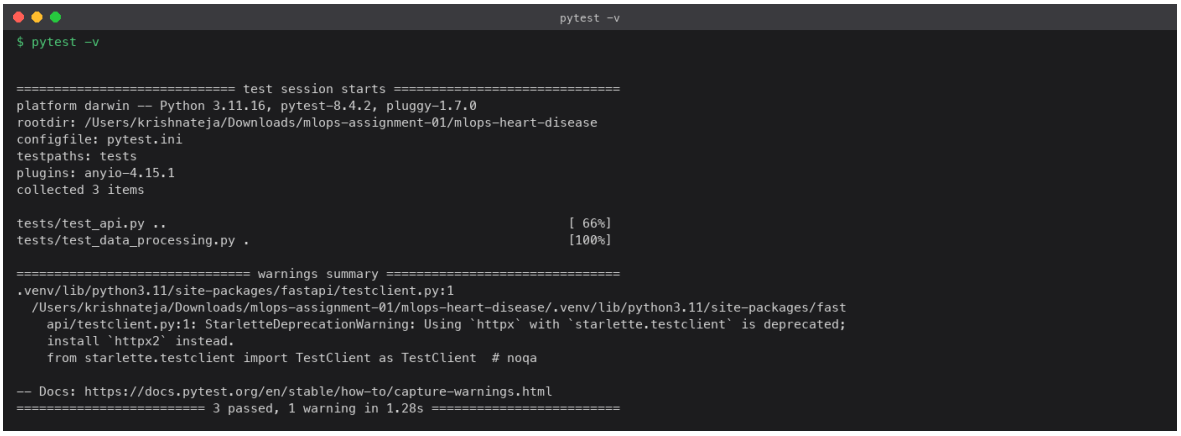
Evidence: 01_eda.png



Evidence: 02_mlflow.png



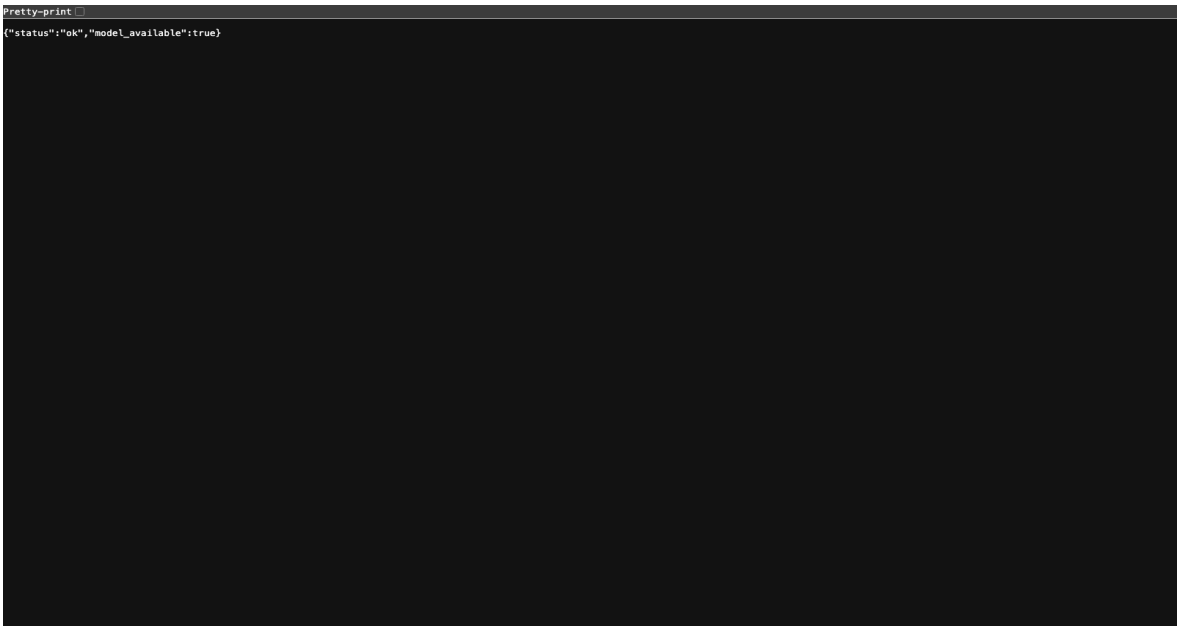
Evidence: 03_tests.png



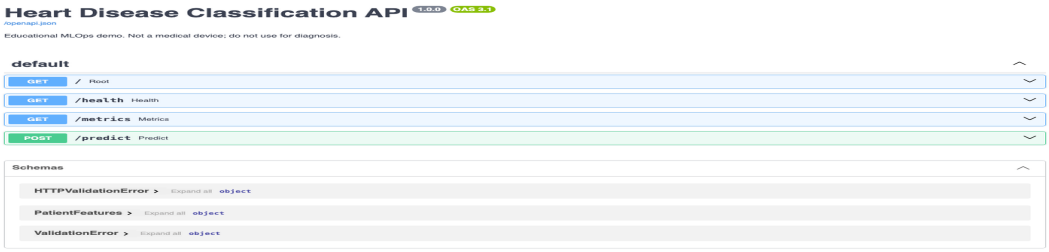
Evidence: 04_lint.png

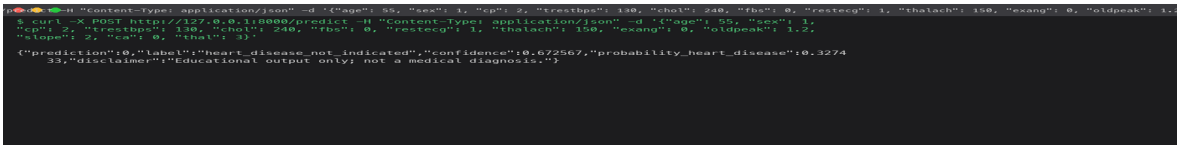


Evidence: 06_api_health.png

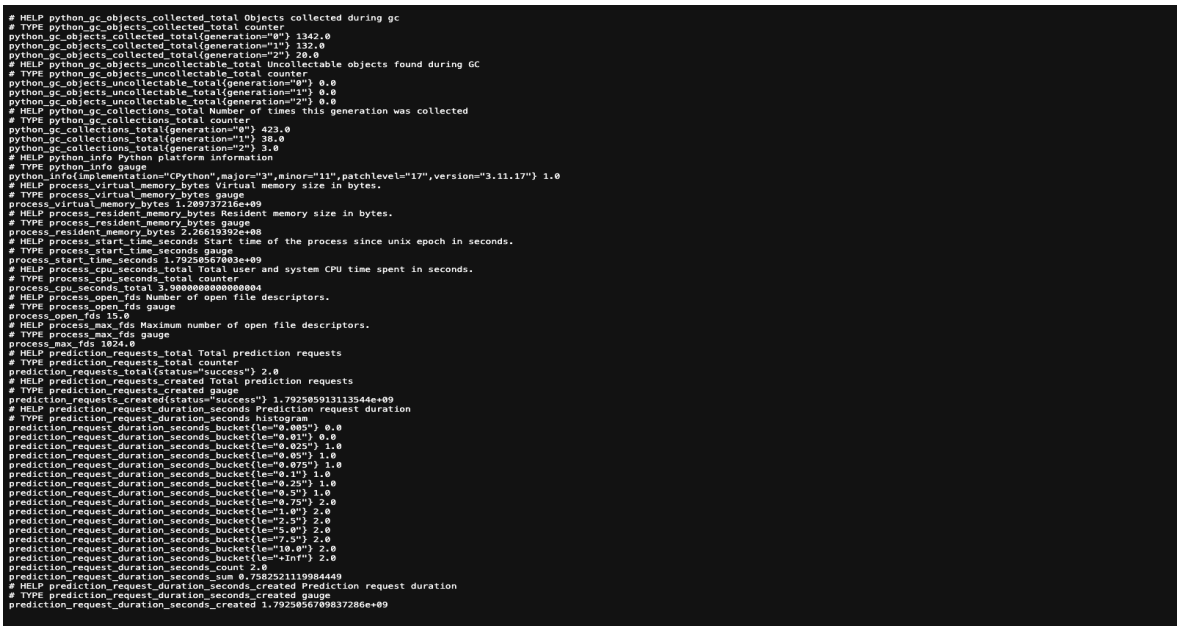


Evidence: 07_api_predict.png





Evidence: 09_prometheus_metrics.png



```
# HELP python_gc_objects_collected_total Objects collected during gc
# TYPE python_gc_objects_collected_total counter
python_gc_objects_collected_total{generation="0"} 1342.0
python_gc_objects_collected_total{generation="1"} 132.0
python_gc_objects_collected_total{generation="2"} 20.0
# HELP python_gc_objects_uncollectable_total Uncollectable objects found during GC
# TYPE python_gc_objects_uncollectable_total counter
python_gc_objects_uncollectable_total{generation="0"} 0.0
python_gc_objects_uncollectable_total{generation="1"} 0.0
python_gc_objects_uncollectable_total{generation="2"} 0.0
# HELP python_gc_collections_total Number of times this generation was collected
# TYPE python_gc_collections_total counter
python_gc_collections_total{generation="0"} 423.0
python_gc_collections_total{generation="1"} 38.0
python_gc_collections_total{generation="2"} 3.0
# HELP python_info Python platform information
# TYPE python_info gauge
python_info{implementation="CPython",major="3",minor="11",patchlevel="17",version="3.11.17"} 1.0
# HELP process_virtual_memory_bytes Virtual memory size in bytes.
# TYPE process_virtual_memory_bytes gauge
process_virtual_memory_bytes 1.280737216e+09
# HELP process_resident_memory_bytes Resident memory size in bytes.
# TYPE process_resident_memory_bytes gauge
process_resident_memory_bytes 2.26619392e+08
# HELP process_start_time_seconds Start time of the process since unix epoch in seconds.
# TYPE process_start_time_seconds gauge
process_start_time_seconds 1.79250567003e+09
# HELP process_cpu_seconds_total Total user and system CPU time spent in seconds.
# TYPE process_cpu_seconds_total counter
process_cpu_seconds_total 5.900000000000004
# HELP process_open_fds Number of open file descriptors.
# TYPE process_open_fds gauge
process_open_fds 15.0
# HELP process_max_fds Maximum number of open file descriptors.
# TYPE process_max_fds gauge
process_max_fds 1024.0
# HELP prediction_requests_total Total prediction requests
# TYPE prediction_requests_total counter
prediction_requests_total{status="success"} 2.0
# HELP prediction_requests_created Total prediction requests
# TYPE prediction_requests_created gauge
prediction_requests_created{status="success"} 1.7925059131135444e+09
# HELP prediction_request_duration_seconds Prediction request duration
# TYPE prediction_request_duration_seconds histogram
prediction_request_duration_seconds_bucket{le="0.005"} 0.0
prediction_request_duration_seconds_bucket{le="0.01"} 0.0
prediction_request_duration_seconds_bucket{le="0.025"} 1.0
prediction_request_duration_seconds_bucket{le="0.05"} 1.0
prediction_request_duration_seconds_bucket{le="0.075"} 1.0
prediction_request_duration_seconds_bucket{le="0.1"} 1.0
prediction_request_duration_seconds_bucket{le="0.25"} 1.0
prediction_request_duration_seconds_bucket{le="0.5"} 1.0
prediction_request_duration_seconds_bucket{le="0.75"} 2.0
prediction_request_duration_seconds_bucket{le="1.0"} 2.0
prediction_request_duration_seconds_bucket{le="2.5"} 2.0
prediction_request_duration_seconds_bucket{le="5.0"} 2.0
prediction_request_duration_seconds_bucket{le="7.5"} 2.0
prediction_request_duration_seconds_bucket{le="10.0"} 2.0
prediction_request_duration_seconds_bucket{le="+Inf"} 2.0
prediction_request_duration_seconds_count 2.0
prediction_request_duration_seconds_sum 0.7582521119984449
# HELP prediction_request_duration_seconds_created Prediction request duration
# TYPE prediction_request_duration_seconds_created gauge
prediction_request_duration_seconds_created 1.7925056709837286e+09
```

11. Conclusion and References

The repository provides an end-to-end implementation scaffold. Final submission should include only results from a genuine local run, verified screenshots, the student's own GitHub repository URL, and a short demo video. Any uncompleted deployment steps should be disclosed rather than represented as successful.

References

1. UCI Machine Learning Repository, Heart Disease, DOI: 10.24432/C52P4X — https://doi.org/10.24432/C52P4X
2. Scikit-learn documentation — https://scikit-learn.org/
3. MLflow documentation — https://mlflow.org/docs/latest/
4. FastAPI documentation — https://fastapi.tiangolo.com/
5. Docker documentation — https://docs.docker.com/
6. Kubernetes documentation — https://kubernetes.io/docs/
7. Prometheus documentation — https://prometheus.io/docs/

Repository URL: add the real URL after pushing to your own GitHub account.